

Raj Mishra

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Profile Summary — Data scientist with 4+ years of experience in NLP, Generative AI, and large-scale ML systems. Proven ability to optimize LLMs for cost and performance, deliver impactful healthcare analytics, and lead cross-functional projects from data engineering to model deployment. Skilled in leveraging state-of-the-art open source and cloud technologies to solve real-world business challenges.

Skills

- **ML Frameworks:** Scikit-learn, XGBoost, Scipy, Numpy
- **Deep Learning:** PyTorch, TensorFlow, Keras, OpenCV
- **Data Engineering:** pySpark, SQL, Airflow, Data Warehouse, Databricks
- **Cloud Platforms:** AWS, Azure, Snowflake
- **NLP & Generative AI:** NLTK, Hugging Face, BERT, GPT
- **Statistical Techniques:** Hypothesis Testing, A/B Testing
- **Languages:** Python, SQL
- **Data Visualization:** Power BI, Plotly, Seaborn
- **Analytics:** Excel, PowerPoint, Story telling

Experience

Ensemble Health Partners

Sep 2023 – Present

Associate II, (AI Engineering)

Next Best Action

- Design: Engineered a scalable NLP pipeline to extract actionable insights from medical notes, And optimized LLM inference by replacing the GPT4 model with GPT 3.5 and open source(Mixtral-8x7B), reducing costs by 90% while maintaining **70% accuracy**.
- Develop a Next-best action system with 20,000+ features, improving the AR decision speed by 40% and covering 65% of accounts, And scaled the Solution from 1 client to 25 clients.
- Built an LLM-based summary model for AR notes and account histories, reducing manual review time by **25-40%** and accelerating follow-up actions for AR agents and increasing resolution rates by 18-25%.

Authorization Denial Prediction

- Collaborated with the innovation team to define the problem statement, establish baseline metrics, and identify critical data gaps for predicting insurance claim denials due to missing authorizations, optimizing revenue cycle management workflows.
- Engineered advanced features from complex, high-dimensional claims data, applying techniques such as Multiple Correspondence Analysis (MCA), Outlier scoring, and SMOTE for class imbalance mitigation, resulting in robust input representations for machine learning models.
- Evaluated multiple supervised learning algorithms, achieving 80% precision at 55% recall-surpassing baseline performance by 50%-to proactively flag high-risk claims prior to submission.
- Designed and deployed scalable batch scoring pipelines using Azure Databricks, and PowerBi dashboard for predictive insights.

BluePi Pvt ltd

May 2021 – Sept 2023

Senior Data Science Engineer

Jubilant Foodworks limited (Dominos, Popyee's)

- Built robust data pipelines and a cloud-based data warehouse (AWS, Snowflake) ensuring high data quality and security. Developed NLP-driven POCs to extract actionable insights from customer feedback, demonstrating the feasibility of machine learning solutions to improve operational efficiency and revenue growth.

Micro Finance Bank

- Developed and deployed machine learning models for credit scoring and fraud detection, reducing loan default rates by 25% and achieving 78% accuracy in identifying fraudulent applications. Leveraged A/B testing and Power BI dashboards to optimize repayment schedules-cutting defaults by 20%-and deliver real-time risk insights for portfolio management.

Education

UIET, Maharshi Dayanand University, Rohtak

Bachelor of Technology in Computer Science and Engineering

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🎥 **Facial Mask Detection:** Developed a computer vision model using deep learning and Computer vision to detect mask-wearing individuals in images, supporting the COVID-19 safety protocols.

Certifications

- **AWS** – Machine Learning Specialist
- **Snowflake** – SnowPro Certification

08/2020 – Present

01/2023 – Present